Prediction of Alzheimer's using Convoluted Neural Networks

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I. INTRODUCTION

Alzheimer's disease (AZM) is a neurological disorder that gradually impairs memory and cognitive abilities, making it difficult to carry out everyday tasks. It typically affects older adults, with an estimated 60 million Americans over the age of 65 having some form of AZM. AZM is ranked as the third leading cause of death in the elderly, after cancer and heart disease. The likelihood of developing AZM increases with age, with about 50% of adults aged 85 or older affected. The disease is the most common form of dementia in older adults[1]. Scientists are still working to understand the complex brain changes that occur in AZM. Although symptoms may not appear until later in life, modifications in the brain begin much earlier. Toxic changes in the brain, such as the buildup of abnormal proteins that create amyloid plaques and tau tangles, occur during the early stages of the disease. AZM primarily affects the limbic system and the entorhinal cortex of the brain, which are critical for forming memories.[2]

The framework for collecting and managing the data for this study on predicting Alzheimer's using CNNs involves the use of publicly available datasets that have been collected and published by various research centers. These datasets contain different types of data, including neuroimaging modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Positron Emission Tomography (PET), genetic and blood information, and clinical and cognitive assessments.

The most commonly used datasets in the field of Alzheimer's research include Alzheimer's Disease Neuroimaging Initiative (ADNI), Australian Imaging, Biomarker,and Lifestyle Flagship Study of Aging (AIBL), Open Access Series of Imaging Studies (OASIS), and Minimal Interval Resonance Imaging in Alzheimer's Disease (MIRIAD). These datasets are publicly available and contain data collected from a large number of participants.

ADNI is notable for being a longitudinal and multicenter study, and it is the most common dataset used in Alzheimer's research. The ADNI study collects data from patients with different stages of cognitive impairment, including normal cognition, mild cognitive impairment (MCI), and Alzheimer's disease. ADNI researchers collect various types of data from patients, including clinical, genetic, MRI, PET images, and biospecimens[3].

AIBL collects imaging and medical data from individuals with AD, individuals with MCI, and healthy individuals without cognitive impairment. OASIS aims to share neuroimaging brain data sets with researchers in related areas. MIRIAD contains MRI scans from AD patients and NC volunteers.

The process of identifying target population involves selecting individuals who may be at risk of the disease, such as those with a family history of Alzheimer's or those who have been diagnosed with mild cognitive impairment (MCI) or Alzheimer's disease. To provide a background of the target population, factors such as age, gender, ethnicity, medical history, and family history of Alzheimer's are considered. Individuals over the age of 65 and women are more likely to develop Alzheimer's disease, and certain ethnic groups, such as African Americans and Hispanics, have a higher risk. The target population for this study also includes individuals with MCI, which involves mild cognitive decline that exceeds what would be expected for their age but does not interfere with daily activities.[4]

II. LITERATURE REVIEW

Ebrahimi, A., Luo, S., and Alzheimer's Disease Neuroimaging Initiative. Convolutional neural networks for Alzheimer's disease detection on MRI images. This article reviews the use of convolutional neural networks (CNNs) for the detection of Alzheimer's disease (AD) on magnetic resonance imaging (MRI) images. The study compares various deep models and configurations, including 2D and 3D CNNs and recurrent neural networks (RNNs), and introduces transfer learning from a dataset of 2D images to 3D CNNs. The results of the study show that the 3D voxel-based method with transfer learning outperforms the other methods with 96.88% accuracy, 100% sensitivity, and 94.12% specificity. The article concludes that CNNs with transfer learning from ImageNet to MRI datasets using 3D CNNs can considerably improve the results of AD detection on MRI scans.The study only focused on detecting AD and did not consider other forms of dementia that may have similar imaging features, such as frontotemporal dementia or Lewy body dementia. Thus, the CNN model may not be effective in detecting these other forms of dementia.[5]

Venugopalan et al. utilized a 3D CNN approach to eliminate noise from MRI scans, extract properties, and focus on the hippocampus brain area for early detection of Alzheimer's disease. The study also utilized PET and neuropsychological assessments, and the ADNI dataset consisted of 558 subjects with 2,794 scans. The dataset included 503 MRI images, 808 SNP, and 2004 HER. They used a three-tier automatic encoder with 199, 99, and 51 nodes for each part

of the Magnetic Resonance scan image, which included 8209 voxels scattered across 18 parts. Their approach involved noise filtering and feature extraction, followed by converting the input data into a 0's and 1's format through short encoding. The study trained and evaluated several models, including a unimodal CNN trained on MRI data, a multimodal CNN trained on all modalities, and a combination of multimodal CNN and GCN for improved feature learning. The results showed that the multimodal CNN and multimodal CNN with GCN outperformed the unimodal CNN, achieving an accuracy of 74% and 76%, respectively, in predicting the Alzheimer's disease stage. The combination of MRI and PET data provided the most significant improvement in accuracy. The study achieved an accuracy of 78% in early detection of Alzheimer's disease using the ADNI dataset. The study demonstrates the effectiveness of deep learning approaches for early detection of Alzheimer's disease using multimodal data, and combining different modalities can improve accuracy. GCN can also enhance feature learning in multimodal models.The study used a relatively small dataset consisting of 558 subjects, which may not be representative of the general population. Therefore, the model's performance may not be generalizable to larger and more diverse datasets.[6]

Mehmood et al. propose a transfer learning approach using MRI images. The study utilizes the ADNI dataset, consisting of 830 subjects with 1728 MRI scans, and involves transfer learning using a pre-trained VGG-16 model, followed by fine-tuning. The proposed method outperforms existing state-of-the-art methods, achieving an accuracy of 95.29%. In contrast, a study by another group uses deep learning with multiple layers called structured deep learning, specifically Visual Geometry Group architecture, on FMRI images gathered from ADNI to detect mild cognitive impairment. The study further classified 300 MRI subjects as Alzheimer's, late mild cognitive, and initial mild cognitive periods using different biomarkers, including structural MRI, PET, and MRI. The techniques used include a multi-layered CNN with various layers, such as convolution layer, pooling layer, and softmax layer. The study achieved an accuracy of 98.73% without data augmentation. Both studies highlight the potential of deep learning approaches in medical imaging applications for early diagnosis of Alzheimer's disease.While both studies achieved high accuracy, it is unclear how the models would perform in a clinical setting. The studies did not explore the interpretability of the model's predictions, which is an essential aspect of clinical decision-making.[7]

Danni Cheng et al.the combination of Convolutional Neural Networks (CNN) and Recurrent Neural Networks were employed in this study.They provide a novel categorization framework constructed on a mix of 2D convolutional neural networks (CNN) and recurrent neural networks (RNN) that studies the characteristics of 3D PET scans by splitting the 3D picture into a series of 2D slices. From the AZM Neuroimaging Initiative (ADNI) database, they tested their technique on basic PET scans from 339 cases, comprising 93 AZM cases,146 mild dementia cases, and 100 normal controls (NC). Experimental results demonstrate that the suggested technique obtains an AUC of 95.28 percent for AZM vs. Normal Control classification and 83.90 percent for Mild Dementia vs. Normal Control classification, indicating categorization performance that is encouraging. Only limited information is used 2D slices of 3D PET scans are trained the proposed framework.[8]

Vijeetha Patil et al propose the superior machine learning approach for the ADNI dataset. The primary focus is on two methods: the 18-layer convolutional network and the 3D convolutional network. It was found that multi-layered CNNs provide more precise outcomes than 3D CNN. Additionally, the study includes a new approach to partitioning the training and testing samples for the ADNI dataset that leads to the highest accuracy achieved with the 18-layer CNN. The main objective of the study is to predict Alzheimer's disease early using machine learning techniques, resulting in a 98% accuracy rate for the 18-layer CNN.[9]

Neuroimaging data is acquired using a variety of imaging modalities, which leads to a lack of uniformity and makes it difficult to compare the outcomes of different research.The study points out that the majority of research are cross-sectional and simply present an overview of the illness. Studies that track participants over an extended period of time may offer more thorough insights into the course of a disease and support the creation of more precise diagnostic models.

There are certain drawbacks to take into account such as variability in the imaging techniques used to acquire neuroimaging data creates a lack of standardization, making it challenging to compare results across studies.These studies are predominantly cross-sectional, providing only a snapshot of the disease. Longitudinal studies, which follow subjects over time, could provide more comprehensive insights into disease progression and aid in developing more accurate diagnostic models.The study highlights the use of limited sample populations in studies, which may not accurately represent the entire population, leading to less effective predictive models for Alzheimer's disease in excluded populations.[10]

Overall the study supports the use of machine learning and deep learning techniques for the diagnosis of Alzheimer's disease using neuroimaging data. These techniques have the potential to provide more accurate and earlier diagnoses, which could lead to improved health outcomes for patients. It suggests that these techniques can be used to identify potential biomarkers for the disease, which could lead to the development of more effective preventative and treatment methods. [10]There are no conflicting findings in the papers examined.

III. CASE STUDY

Our case study involves the use of CNN+Softmax for the prediction of AZM.

Datasets

Dataset taken for this case study is Sarvesh Dubey's public dataset from the kaggle database for the purpose of training and testing the model. This dataset accommodates 6400 MRI images of the encephalon. It is categorized into four different classes of AZM which includes, non-demented (no indication of Alzheimer's), mildly demented, moderately demented, and very mildly demented.

The study uses MRI scans of patients with Alzheimer's disease and applies a CNN architecture to extract important features from the scans. The CNN architecture includes convolutional layers, activation functions, and max-pooling operations, which help to identify patterns in the data. A dense layer with Softmax classifier is used to classify the images into four different groups: mild demented (MD), moderate demented (MOD), non-demented (ND), and very mild demented (VMD). The performance of the proposed CNN+Softmax model is then compared with a CNN+SVM model using evaluation metrics such as accuracy, precision, recall, and f1-score.

A. Architecture of CNN Softmax

The performance of a CNN mainly depends on the architecture of the layers and the filters used in the convolutional layers.

The proposed CNN+Softmax model is a deep learning architecture that is designed to classify AZM MRI images into four categories: mild demented (MD), moderate demented (MOD), non-demented (ND), and very mild demented (VMD). The architecture consists of several layers that extract features from the input MRI images and make predictions using the Softmax classifier.

The first layer (L1) has two Convolution operation (Conv2D) with 32 filters and 3x3 kernel size, ReLU activation function, Batch normalization, as well as Max-pooling operation with 2x2 pool size and strides of (2, 2). The input shape of the MRI in this layer is 150x150x3.

The second layer (L2) has one Conv2D operation with 64 filters and 3x3 kernel size, a Rectified Linear Unit activation layer, Batch normalization layer, and Max-pooling operation with 2x2 pool size and strides of (2,2).

The third layer (L3) is similar to layer 2.

The fourth layer (L4) is a Flatten layer that reshapes the data from the last layer into a single column of data.

The fifth layer (L5) is a Fully Connected layer (dense layer) with 320 neurons and a Rectified Linear Unit activation function.

The last layer, i.e., layer 6 (L6), contains a fully connected layer with 4 output neurons for each class of AZM, and the Softmax function, which classifies the AZM MRI scans into 4 categories, namely MD, MOD, ND, VMD. The Softmax function scales the predicted probabilities so that they sum to 1, and returns them as the final classification result

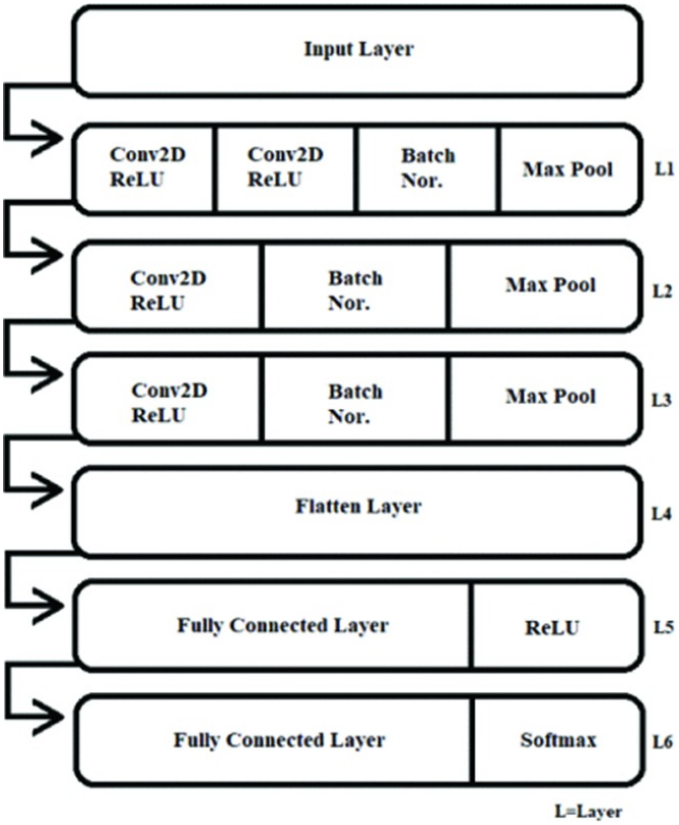


Fig. 1. CNN+Softmax model architecture.

B. Comparison of CNN+SVM with CNN+Softmax

The two popular architectures used for image classification, namely the CNN+SVM and CNN+Softmax.The CNN+SVM model is formed by replacing the fully connected layer (dense layer) of CNN+Softmax model with SVM classifier. CNN+Softmax model is trained to output probabilities for each class, which can be interpreted as the likelihood that the input belongs to that class. The softmax function ensures that the probabilities sum to one, making it easier to compare the relative probabilities for each class.

The SVM classifier, which is trained to find the hyperplane that maximally separates the classes. The SVM is a binary classifier, so it needs to be extended to handle multi-class classification by using techniques such as one-vs-all or one-vs-one.

In terms of accuracy, both models can achieve high performance on the task when they are properly trained and tuned. However, the CNN+Softmax model is more suitable for problems where the decision boundary between classes is not well-defined, since it can output probabilities for each class that reflect the degree of uncertainty.

Strengths

The CNN+softmax architecture used for predicting Alzheimer's disease has several advantages and limitations. One of the main advantages of the model is its high accuracy in image classification tasks. The model can accurately classify brain images into normal and Alzheimer's classes, making it a useful tool for diagnosis. Additionally, the CNN+softmax model is an end-to-end learning architecture that can learn features from raw data, allowing the model to automatically learn the important features from input images without any manual feature engineering. The model's parallel processing also allows for faster execution, and transfer learning can be applied to the model, making it useful for smaller datasets.

Limitations

However, there are also limitations to the CNN+softmax architecture. One of the main limitations is its limited interpretability, as the model is a black-box model that can be difficult to interpret how it is making predictions. This can be a limitation in medical applications where interpretability is important for understanding the underlying pathology. The CNN+softmax model also requires a large amount of labeled data to achieve high accuracy, which can be challenging to obtain in medical applications. Overfitting is another potential limitation, where the model may learn the training data too well and perform poorly on unseen data. Finally, the CNN+softmax model may not generalize well to new unseen data, particularly if the new data is significantly different from the training data.

IV. Results and Discussion

Implementing a CNN architecture for predicting Alzheimer's disease would come with several cost implications that need to be considered. Firstly, there would be costs related to acquiring and maintaining the hardware and software infrastructure needed to run the CNN model. This may include high-performance computing systems with specialized GPUs to accelerate the training and prediction processes, as well as ongoing maintenance costs and electricity costs. Secondly, there would be costs associated with acquiring and processing the MRI data required to train and test the CNN model. This could include the cost of MRI scans, as well as additional costs for preprocessing and cleaning the data. Thirdly, there would be costs associated with training and optimizing the CNN model, such as hiring experienced data scientists and machine learning experts to design, develop, and train the model. Moreover, there may be indirect costs, such as developing and implementing policies and procedures for integrating the model into clinical practice and training healthcare providers on how to use and interpret the model's outputs. It is important to factor in these costs to ensure the successful implementation of the CNN architecture for Alzheimer's prediction.

The use of Convolutional Neural Networks (CNNs) for the prediction of Alzheimer's disease can have a significant positive impact on population health management.With accurate prediction, healthcare providers can identify patients with early signs of Alzheimer's and intervene early, improving outcomes and potentially reducing healthcare costs. Early detection and intervention can lead to better management of symptoms, reduced hospitalizations, and improved quality of life for patients and their caregivers. This can ultimately result in better patient outcomes and reduced healthcare costs.

The use of CNNs for Alzheimer's prediction can help healthcare providers prioritize patients for further evaluation, reduce unnecessary testing and imaging, and improve resource allocation. This can be especially valuable in settings where resources are limited, and the demand for diagnostic tests is high. By prioritizing patients for evaluation, healthcare providers can ensure that resources are used efficiently and effectively.

The use of CNNs for Alzheimer's prediction can aid in research and development efforts aimed at finding better treatments and ultimately a cure for the disease. Accurate prediction and early intervention can lead to better identification of potential participants for clinical trials, improving research outcomes and accelerating the development of new treatments.

Alzheimer's prediction can also have a significant impact on reducing the burden on caregivers. Early detection and intervention can lead to better management of symptoms, reducing the need for extensive and often costly care from family members and caregivers. Furthermore, the use of CNNs for Alzheimer's prediction can also help identify patients who may benefit from support services, such as counseling, respite care, and caregiver training. By providing caregivers with the resources they need to provide effective care, the burden on caregivers can be reduced, improving their quality of life and reducing the risk of caregiver burnout.

Overall, the use of CNNs for Alzheimer's prediction can have a significant impact on population health management by improving outcomes, reducing costs, and accelerating research efforts. By accurately predicting the onset of Alzheimer's disease, healthcare providers can intervene early, improving patient outcomes and reducing the burden on the healthcare system.

V. Conclusion

It is clear from the findings of the several research mentioned in this article that deep learning approaches, particularly CNNs and RNNs, have shown great potential for the early detection and diagnosis of Alzheimer's disease using medical imaging data. These approaches have demonstrated high accuracy in detecting Alzheimer's disease and mild cognitive impairment, but their interpretability and generalizability need further exploration.The proposed CNN+Softmax model for predicting different types of AZM

from MRI scans of the brain. The model achieved a high accuracy of 98.59% and outperformed the CNN+SVM model on various evaluation metrics. The study suggests that further improvements can be made by adding a color balancing parameter, using the Probabilistic Neural Network (PNN) approach for result optimization, and increasing the model's learning capacity. The study highlights the potential of using deep learning models for predicting AZM from MRI scans, and further research is needed to evaluate the models' performance in a clinical setting and enhance their interpretability and generalizability.

Classes	Precision	Recall	F1-Score	Support
Mild Demented	93%	82%	87%	171
Moderate Demented	100%	94%	97%	16
Non Demented	91%	96%	94%	640
Very Mild Demented	91%	88%	90%	455

Fig. 2. Performance of CNN+SVM.

Classes	Precision	Recall	F1-Score	Support
Mild Demented	98%	98%	98%	171
Moderate Demented	100%	88%	93%	16
Non Demented	99%	99%	99%	640
Very Mild Demented	99%	99%	99%	455

Fig. 3. Performance of CNN+Softmax.

VI. Recommendations

To improve the performance of a CNN model in detecting Alzheimer's disease certain recommendations can be implemented. First, a larger dataset can be gathered by collaborating with more medical institutions to acquire more MRI scans. Second, addressing class imbalance by increasing the number of samples for underrepresented classes can be done through data augmentation or collecting more data. Third, fine-tuning hyperparameters such as learning rate, batch size, and number of epochs can improve model performance. Fourth, leveraging pre-trained CNN models through transfer learning from ImageNet can save time and resources. Fifth, incorporating clinical data such as age, gender, and cognitive test scores as additional features can potentially improve model performance. Lastly, techniques such as Grad-CAM can be used to improve model interpretability, enabling clinicians to identify the most important regions of the brain for classification.

Feasible recommendations: Several suggestions can be taken into account to enhance the performance of a CNN+Softmax model for the categorization of Alzheimer's disease. First, we may increase the dataset by gathering more MRI images of Alzheimer's patients and healthy people, which will give the model access to a wider range of training data. However, this can be difficult as it is dependent on ethical considerations and patient privacy. Additionally, the model's performance may be improved by tuning its hyperparameters, like the learning rate, batch size, and number of epochs. Furthermore, enhancing the input data's quality by preprocessing the pictures with techniques like normalization or augmentation could help in lowering noise and artifacts that can adversely affect the model's accuracy. Fourthly, adding a color balancing parameter to the CNN model can increase the classification accuracy of textured MRI encephalon pixels. Finally, implementing transfer learning by incorporating pre-trained models can improve the prediction accuracy, but it requires careful consideration of the similarities between the pre-trained model's architecture and the target dataset.

However there are also some challenges, despite the fact that these suggestions have the ability to considerably enhance the model's performance. Due to ethical and privacy considerations, obtaining additional patient data can be difficult, and gathering data from multiple domains could require significant coordination and resources. Additionally, it can be challenging to collect data from diverse age groups because doing so frequently necessitates lengthy, resource-intensive investigations.

VII. References

1)"Alzheimer's Disease Facts and Figures." Alzheimer's Association, www.alz.org/alzheimers-dementia/facts-figures.

2)National Institute on Aging. "What Happens To The Brain In Alzheimer's Disease? CAUSES OF ALZHEIMER'S DISEASE." U.S. Department of Health and Human Services, 28 Oct. 2021, www.nia.nih.gov/health/what-happens-brain-alzheimers-disease.

3)Syaifullah, A. H., Shiino, A., Kitahara, H., Ito, R., Ishida, M., Tanigaki, K. (2021). Machine Learning for Diagnosis of AD and Prediction of MCI Progression From Brain MRI Using Brain Anatomical Analysis Using Diffeomorphic Deformation. Alzheimer's and Dementia: Diagnosis, Assessment and Disease Monitoring, 13(1), e12216. doi: 10.1002/dad2.12216 https://www.frontiersin.org/articles/10.3389/fneur.2020.576029/full

4)Alzheimers society Risk factors for Alzheimer's disease https://www.alzheimers.org.uk/about-dementia/types-dementia/who-gets-alzheimers-disease

5) Ebrahimi, A., Luo, S., and Alzheimer's Disease Neuroimaging Initiative. (2019). Convolutional neural networks for Alzheimer's disease detection on MRI images. Journal of healthcare engineering, 2019. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8083897/

6) Venugopalan J, Tong L, Hassanzadeh HR. Multimodal deep learning models for Early Detection of Alzheimer's disease stage. Sci Rep. 2021. https://doi.org/10.1038/s41598-020-74399-w.

7) Mehmood A, Yang S, Feng Z, Wang M, Ahmad AS, Khan R, et al. A transfer learning approach for early diagnosis of Alzheimer's disease on MRI images. Neuroscience. 2021;460:43–52. https://doi.org/10.1016/j.neuroscience.2021.01.002.

8) D. Cheng and M. Liu, "Combining convolutional and recurrent neural networks for Alzheimer's disease diagnosis using PET images," IST 2017 - IEEE Int. Conf. Imaging Syst. Tech. Proc., vol. 2018-Janua, no. Mci, pp. 1–5, 2017, doi: 10.1109/IST.2017.8261461. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9954686/

9) Patil, V., Madgi, M., Kiran, A. (2021). Early prediction of Alzheimer's disease using convolutional neural network: a review. Journal of Ambient Intelligence and Humanized Computing, 12(10), 11191-11204. https://ejnpn.springeropen.com/articles/10.1186/s41983-022-00571-w

10) Zhao, Z., Chuah, J. H., Lai, K. W., Chow, C. O., Gochoo, M., Dhanalakshmi, S., Wang, N., Bao, W., Wu, X. (2021). Conventional machine learning and deep learning in Alzheimer's disease diagnosis using neuroimaging: A review. Frontiers in aging neuroscience, 13, 657155. doi: 10.3389/fnagi.2021.657155 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9939698/

11) Singh, P Mishra, S. K.(2020).Alzheimer's detection and categorization using a deep-learning approach. Journal of Ambient Intelligence and Humanized Computing, 11(12), 5535-5550. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=arnumber=9917774

12) Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: a simple way to prevent neural networks from overfitting. J. Mach. Learn. Res. 15, 1929–1958. https://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf

13) Lin, W.,Tong,T., Gao,Q.,Guo, D. (2018). Convolutional Neural Networks-Based MRI Image Analysis for the Alzheimer's Disease Prediction From Mild Cognitive Impairment. Frontiers in Neuroscience, 12, 777. doi: 10.3389/fnins.2018.00777. https://www.frontiersin.org/articles/10.3389/fnins.2018.00777/full